

# earrings.ai

Yosun Chang  
yc@AIMagical.com  
AIMagical Permute.xyz  
San Francisco, CA, USA

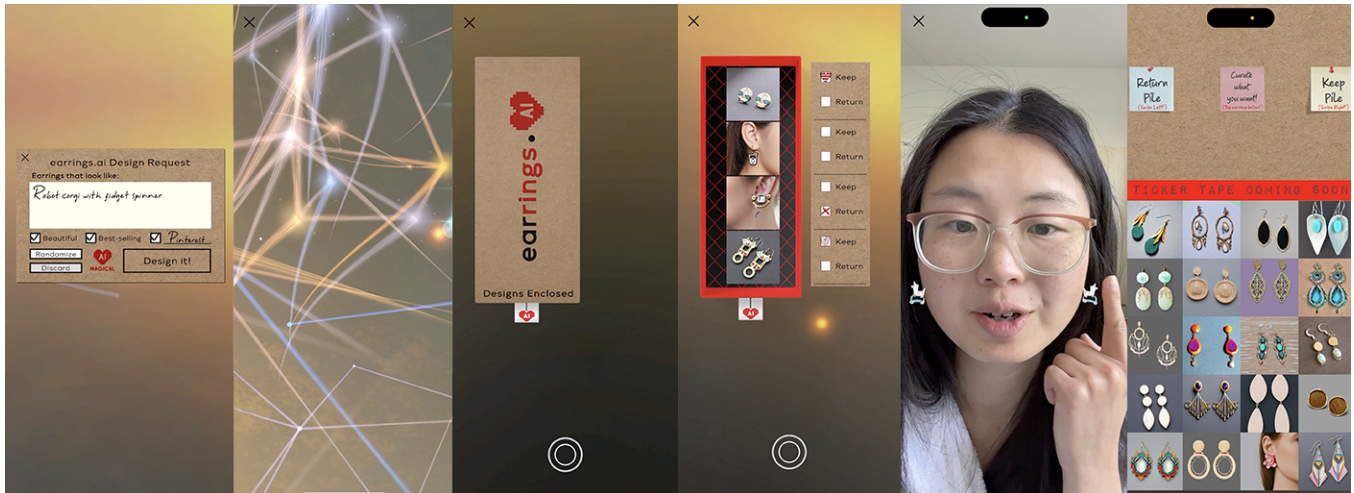


Figure 1: Seattle Mariners at Spring Training, 2010.

## ABSTRACT

earrings.ai is an AI design app platform that lets you create your dream earrings simply from describing it – and AR try them on [2] and even participate in a virtual and real world marketplace [1] to have artisan creators make them, or a network of 3D printers fab them!

## CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; **Collaborative interaction**; **Graphical user interfaces**; **Natural language interfaces**; **HCI theory, concepts and models**; **Pointing**; **Text input**; **Gestural input**; **Auditory feedback**; **Touch screens**; **Ubiquitous and mobile computing**; **Ubiquitous computing**; **Mobile computing**; **Ambient intelligence**; **Smartphones**; **Interactive whiteboards**; **Tablet computers**; **Mobile devices**; **Mobile phones**; **Ubiquitous and mobile computing systems and tools**; **Collaborative and social computing**; **Human computer interaction (HCI)**; **Interaction paradigms**; **Mixed / augmented reality**; **Collaborative interaction**; **Natural language interfaces**; **Graphical user interfaces**; **Command line interfaces**; **Empirical studies in HCI**; • **Software and its engineering**; • **Information**

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systems → *Data management systems*; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Arts and humanities**; **Electronic commerce**;

## KEYWORDS

AI, Manufacturing Pipeline, Marketplaces, Social Commerce

### ACM Reference Format:

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## 1 INTRODUCTION

earrings.ai is the first iOS and Android app that is significantly programmed by AI, that enables anyone to create their own AI-generated original earrings simply from describing it in words. The app lets you take your idea from words to product image to 3D AR preview to a virtual tradeable or real world manufactured product. Being part of the indie SIGGRAPH Appy Hour track, the app is notably built by one person (with a budget of 0) and a team of AI. Significant portions, such as the Candy Crush interactivity in browsing earrings, are 99% programmed by AI. The human defines the app and its architecture (managing the AI to program it), and the rich interactions that make the user experience memorable and fun.

## 2 V1 IMPLEMENTATION

The app features Curate, Create, and Collect mode. The Curate mode helps the system understand the user's preferred aesthetics,

which guides the output generated in the Create mode, from just showing stable diffusion results. Collect mode is where we add market dynamics to the mix.

## 2.1 Curate Mode

Curate establishes subtle aesthetic parameters in your jewelry tastes, based on what you select, keep or return.

Curate mode utilizes hybrid catchy UX – both the Tinder swipe UX as well as a bouncy Candy-Crush like UI animation when you select an item to be curated.

## 2.2 Create Mode

Create mode lets you “create your dream earrings, with a starting prompt of “earrings that look like:”, on a skeuomorphic cardboard print “Design Request” form, that you compose. You can dropdown-checkbox some common requests like, “best selling,” etsy, fashion week etc.

Create Mode, in lieu of a traditional loading bar, lets you wave your phone around like a magic wand, as you wait for generation.

## 2.3 Collect Mode

Collect mode lets you review your creations into collections and also collect others and explore both a virtual goods and AI to real world goods Marketplace.

## 3 AESTHETICS

The user experience is further enhanced by a hybrid Maker-style scrapbooking aesthetic.

Cardboard UIs with virtual earrings arriving via cardboard boxes, to be environmentally friendly!

## 4 GEN AI REAL WORLD PRODUCT CREATION

We connect AI to the real world by enabling a pipeline for AI created material goods - on demand, by the human.

### 4.1 Creating real products at the speed of thought

We turn your idea, expressed as natural language words, into real products, currently, using a finetuned version of stable diffusion 1.5 and user-directed aesthetic matching, from what the users chooses to Curate.

### 4.2 Infinite products for AR previews

The images are then processed in a pipeline that turns them into 3D models for AR preview or 3D printing or as blueprints for human artists to create hand made interpretations of.

### 4.3 Marketplace of AI Designs and Open Bidding by Human Artisans

Because these are real world tangible products being created by AI, the creator is potential the client and buyer as well, if he or she chooses to have artisan jewelers on the marketplace make the real version of his/her design.

The designs are placed on a marketplace where artisan artists can bid to create the real versions of for the potential creator-buyer-client. Alternatively, any design can be licensed.

## 4.4 Collectibles and Artificial Scarcity

In addition, we experiment with applying social psychological insights in artificial scarcity in Collectibles mode, to further finetune our model.

## 5 1 HUMAN + TEAM OF AI

Earrings.ai is an experiment in both building an entire full platform app and business with just 1 human (+ 1 corgi) + 1 AI.

### 5.1 Focus for the Human

Unique UI and quirky ideas for a straight-laced team of AI programmers to implement.

### 5.2 Pipelines to delegate to AI

I have also created several pipelines to make such delegation more efficient.

### 5.3 Using AI for what AI is currently good at

When earrings.ai was architected in November 2022, AI could sometimes be relied upon for creating some Unity plugins. The UI and Candy Crush animation code for Curate mode was thus delegated to AI to code in January 2022.

In March 2023, Collect mode was architected and delegated to AI.

As of May 2023, the entire marketplace (Unity client side and serverside) has been delegated to AI to create.

### 5.4 Efficiency and Information Entropy

AI, when given sufficient guidelines on common programming tasks, is able to perform flawlessly and autonomously but sometimes runs into inefficiencies that resemble human “brain farts.” AI can usually be instructed to loop through these to get to unit test passing phases. Except when it gets truly stuck, almost like a precocious child who can function alone but pulls a tantrum to acquire human attention. In which case, a human with significant intuition can change the prompt up for AI to generate the correct result - generally, when instructed to change its prompt, the AI seems unable to do this autonomously to avoid the vicious loops.

I wonder if this has something to do with information entropy  $\Delta S$  and spooky fabric of reality chaos theory laws on maximal efficiency rates. We will somehow still need to add in the human?

## 6 CONCLUSION

Finetuning the final portions with AI.. in an open beta with real human testers... I am about to launch the app with a team of AI marketers. Stay tuned!

## REFERENCES

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